

Lecture Notes on Differential Equations and Transform Techniques

Lecture Notes for Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques

Lecture Notes - 08

Topics covered :

1. Implicit Methods for Numerical Solution of Differential Equations
2. Newton's method for finding roots of a nonlinear equation

- Consider the initial value problem

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \quad (1)$$

The Backward Euler method approximates the solution to the IVP by discretizing the solution over small time steps and using the tangent slopes at the **next step, rather than the current one**. This contrasts with explicit methods, such as the Forward Euler or second-order Runge-Kutta methods, which use known values at each step to update the solution.

Given y_0 at x_0 , the goal is to approximate the solution at the next time step $x_1 = x_0 + h$, where h is the step size. In the Backward Euler method, the key idea is to use the slope of the solution at the new point, (x_1, y_1) , to update our approximation (see Fig. 1).

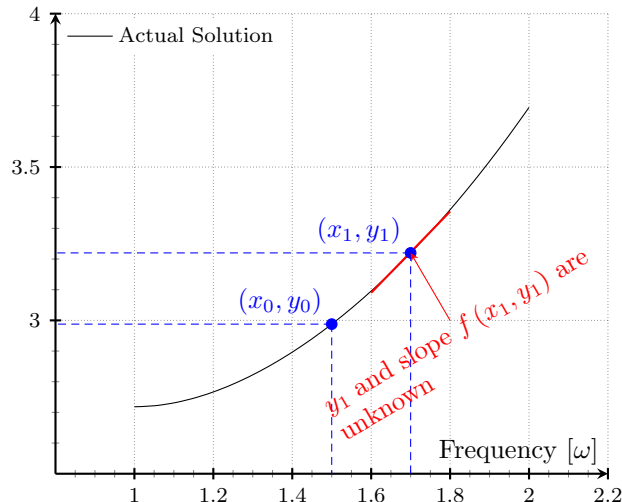


Fig. 1: Illustration of the Backward Euler method

This is different from the explicit methods, where the slope at the current point is used to project forward. In contrast, the Backward Euler method is **implicit** because it requires us to evaluate the slope at the next unknown point. The update equation for y_1 is

$$y_1 = y_0 + hf(x_1, y_1) \quad (2)$$

Since y_1 is not known, (2) forms an implicit relationship. Thus, in order to compute y_1 , it is necessary to adopt numerical methods such as **Newton's method**. This can be computationally intensive, particularly when the

function $f(x, y)$ is non-linear. Once y_1 is computed, it can be used as a starting point for the next step, and the process continues.

The general update equation for the backward Euler method is

$$y_{n+1} = y_n + hf(x_{n+1}, y_{n+1}) \quad (3)$$

- The **Trapezoidal method** for numerically solving the IVP that improves upon the explicit methods by incorporating information (including slope) at both the current and the next time step.

Given y_0 at x_0 , the goal is to approximate the solution at the next time step $x_1 = x_0 + h$, where h is the step size. In the Trapezoidal method, the key idea is to use the average slope of the solution at the new point and the current point to update our approximation (see Fig. 2). Similar to the **Backward Euler method**, the **Trapezoidal method**

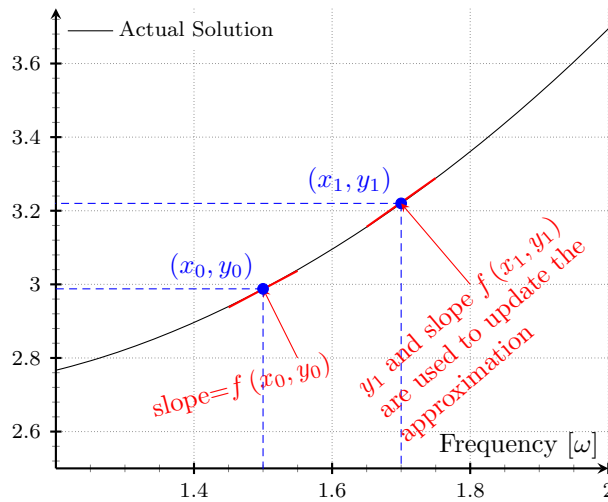


Fig. 2: Illustration of the Trapezoidal method.

is also **implicit** because it requires us to evaluate the slope at the next unknown point. The update equation for y_1 is

$$y_1 = y_0 + h \left[\frac{f(x_0, y_0) + f(x_1, y_1)}{2} \right] \quad (4)$$

Here, y_1 appears on both sides of the equation, making it an implicit method. This means that we cannot directly compute y_1 without solving for it iteratively. Typically, numerical techniques such as **Newton's method** are used to solve for y_1 . Once y_1 is computed, it can be used as the starting point for the next time step, and the process continues.

The general update equation for the Trapezoidal method is

$$y_{n+1} = y_n + h \left[\frac{f(x_n, y_n) + f(x_{n+1}, y_{n+1})}{2} \right] \quad (5)$$

- **Example:** Derive the general update equations for the Backward Euler method and Trapezoidal method for solving

the IVP [1]

$$\frac{dy}{dx} = x + y \quad (6)$$

The general update equation for the Backward Euler method is

$$y_{n+1} = y_n + hf(x_{n+1}, y_{n+1}) \quad (7)$$

Substituting $f(x, y) = x + y$ and $x_{n+1} = x_n + h$ gives

$$y_{n+1} = y_n + h(x_n + h + y_{n+1}) \quad (8)$$

Rearranging to isolate y_{n+1} gives

$$y_{n+1}(1 - h) = y_n + h(x_n + h) \quad (9)$$

Thus, the update equation for Backward Euler is

$$y_{n+1} = \frac{y_n + h(x_n + h)}{1 - h} \quad (10)$$

The general update equation for the Trapezoidal method is

$$y_{n+1} = y_n + \frac{h}{2} (f(x_n, y_n) + f(x_{n+1}, y_{n+1})) \quad (11)$$

Substituting $f(x, y) = x + y$ and $x_{n+1} = x_n + h$ gives

$$y_{n+1} = y_n + \frac{h}{2} (x_n + y_n + x_n + h + y_{n+1}) \quad (12)$$

Rearranging to isolate y_{n+1} gives

$$y_{n+1} \left(1 - \frac{h}{2}\right) = y_n + \frac{h}{2} (2x_n + y_n + h) \quad (13)$$

Thus, the update equation for the Trapezoidal method is:

$$y_{n+1} = \frac{y_n + \frac{h}{2} (2x_n + y_n + h)}{1 - \frac{h}{2}} \quad (14)$$

- Consider a problem of finding atleast one root of the equation $g(z)$ i.e., the problem of solving $g(z) = 0$. One way to find the roots of the function $g(z) = 0$ is by using Newton Raphson method. In Newton Raphson method, starting at an initial point (say z_0), an iterative process is adopted wherein at each iteration, an estimate to the root is computed by finding the roots of the linear approximation of the function (at the previous estimate). The iterative process is repeated till the convergence criteria (i.e., $|g(z_k)| < \epsilon$) is satisfied.
- Let z_k denote the trial point (for root) or estimate of the root at iteration k . The linear approximation of $g(z)$

around z_k is given by

$$l(z) = g(z_k) + (z - z_k)g'(z_k) \quad (15)$$

The root (say z_+) of the linear approximation (i.e., the value of z that satisfies $l(z) = 0$) is given by

$$l(z) = 0 \implies g(z_k) + (z - z_k)g'(z_k) = 0 \implies z^+ = z_k - \frac{g(z_k)}{g'(z_k)} \quad (16)$$

Geometrically, the root z_+ of the polynomial $l(z) = 0$ is the point where the tangent to the curve $g(z) = 0$ at z_k meets the x -axis (see Fig. 3). The updated trial point or estimate for the root of $g(z)$ is

$$z_{k+1} = z_k - \frac{g(z_k)}{g'(z_k)} \quad (17)$$

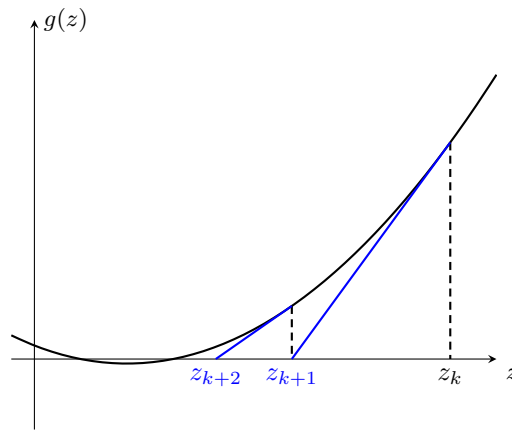


Fig. 3: Illustration of Newton Raphson Method

- The steps involved in Newton Raphson method are outlined in Algorithm 1.

Algorithm 1 Newton Raphson method for Solving $g(x) = 0$

- 1: Choose z_0 and let $k \leftarrow 0$.
 - 2: **while** $|g(z_k)| > \epsilon$ **do**
 - 3: $z_{k+1} = z_k - \frac{g(z_k)}{g'(z_k)}$
 - 4: $k \leftarrow k + 1$
 - 5: **end while**
-

References

- [1] C. H. Edwards and D. E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th ed. Upper Saddle River, NJ: Pearson, 2000.